Gate 1 — Specs and backlog

Oct 29, 2026 · @Jeremy Artero

The gate 1 web prototype takes about 52 hours to build and answers three questions, asked of 20 to 30 warm testers: is the daily heist fun, do people share it, and do they come back the next day? These tests can only kill the project; only gate 2, with at least 300 cold players, authorizes building the MVP.

Framing

Gate 1 looks for a reason to stop before committing to the MVP: it validates nothing, it can only kill the project or correct it.

Five indicators, three questions

Question	Indicator (defined in "Metrics")	Kill or fix if
Is the game understood?	Clarity: T1 cleared without help in under 60 s	Below 60%: fix the onboarding, not the concept
Is it fun?	Voluntary play: 8 minutes of active play on day one, or tutorial and daily heist completed	Below 40%: pivot to Daily Reflect if the feel is good, otherwise one 10-hour iteration on the feel
Is it fun?	Retry after failure	Fewer than 60% of failures followed by a new attempt
Do people share it?	Spontaneous sharing	Below 10%: the viral thesis is dead
Do people come back?	Next-day return without a reminder	Below 30%, even among friends

Succeeding on all five proves nothing: it only unlocks the cold test of gate 2.

Budget and schedule

- **Time:** 52 hours (detail in the backlog), including 8.5 hours of feel sessions and testing, plus 1.75 hours of optional items to cut first.
- **Money:** $0 to $50: a domain name, analytics on a free tier, royalty-free sounds.

- **Schedule:** about 3 weeks of evenings to build, 1 week of testing, decision on day 8.

In scope

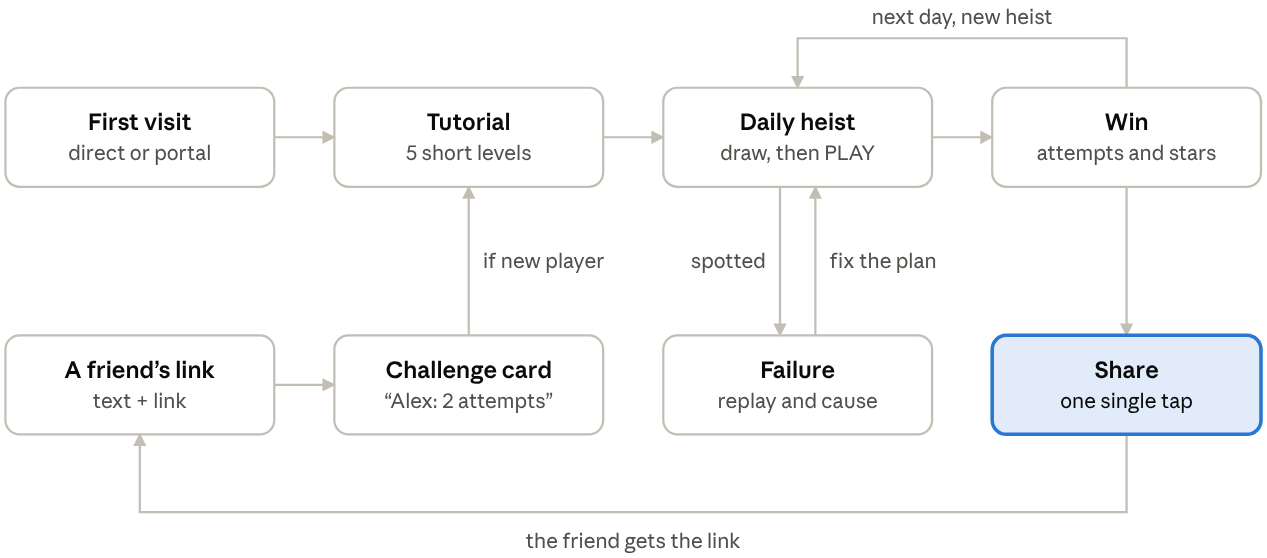
- Web prototype, designed for mobile in portrait, playable on desktop.
- 5 tutorial levels and 7 daily heists, one per test day, including 3 hand-made "signature" heists.
- One-tap sharing and challenge links, with no application server.
- Daily loop: daily heist, streak, countdown.
- Measurement: events, consent, dashboard of the five indicators.
- French and English.

Out of scope

Duel editor, practice mode, ads, purchases, accounts, leaderboards, notifications, iOS and Android apps, doors and switches, application server. Any addition goes through the product owner (Jeremy).

Player journey

Two entry points, one loop: clear, share, come back the next day



Player journey · 2 entry points, 8 steps, 3 loops

A new player lands with no title screen and learns through five short levels. A player arriving from a link first sees the challenge card, then the three-level express tutorial if they have never played; a returning player goes straight to the heist. A failure brings the player back to their intact plan, a win leads to sharing, and the shared link opens the same loop for a friend.

Game rules

The game is a deterministic simulation on a grid: the same plan always produces the same result. Every failure can therefore be explained, and every level can be verified by the solver.

Elements

Element	Rule
Grid	8 columns × 12 rows, portrait. Coordinates (x, y), origin at the top left. Cell types: floor, wall, start, diamond, exit.
Time	One tick = one step of one cell. Execution shown at 4 ticks per second, with an optional ×2.
Thief	Follows the player's plan: one orthogonal step per tick, or a wait. Cannot pass through walls.
Plan	A sequence of adjacent cells drawn with a finger, plus one-tick pauses (3 at most by default, configurable per level). It goes through the diamond and ends on the exit; 64 steps at most, pauses included. No immediate U-turn: moving back onto the previous cell erases the last step.
Guards	Follow a fixed patrol, one step per tick, either as a loop or back and forth. A guard always faces its next cell; during a wait it keeps its facing or takes the one set by the level. Vision cone range 3, configurable.
Cameras	Mounted on a wall. Rotate between 1 and 4 directions and stay `hold` ticks on each. Same cone as guards.

Element	Rule
Lasers	A segment of floor cells, on for `on` ticks then off for `off` ticks, shifted by `phase`.
Periods	Every patrol and cycle lasts a number of ticks that divides 24 (1, 2, 3, 4, 6, 8, 12 or 24). The world therefore repeats at least every 24 ticks, which keeps the solver small.

Vision cone

A guard facing right with range 3 sees the cells (dx, dy) such that 1 ⩽ dx ⩽ 3 and |dy| ⩽ dx − 1, i.e. 9 cells:

```
          dx = 1  2  3
dy = -2         .  .  x
dy = -1         .  x  x
dy =  0   G     x  x  x
dy = +1         .  x  x
dy = +2         .  .  x
```

Other facings are rotations of this shape. A cone cell is seen if it is inside the grid, is not a wall, and no wall cuts the Bresenham line between the guard and that cell, endpoints excluded. A guard sees neither its own cell, nor its sides, nor behind it: sneaking past its back is allowed, and intended.

Tick resolution order

Going from tick t to tick t + 1:

1. The thief executes step t of its plan: a move or a wait.
2. Each guard advances one step along its patrol; its facing is recomputed.
3. Cameras and lasers take their state for tick t + 1.
4. Detection, in this order: contact (the thief is on a guard's cell, or they just swapped cells); vision by a guard; vision by a camera; an active laser under the thief. The first rule triggered gives the cause: guard, camera or laser.
5. If the thief is not caught, it picks up the diamond when on its cell; if it is on the exit with the diamond, the level is won.

At tick 0, nobody sees the start cell: the validator checks it. Being seen on the exit counts as a capture.

Score, stars and attempts

- **Ticks:** duration of the winning plan, pauses included.
- **Par:** the minimum number of ticks found by the solver, shown from the start of the level.
- **Stars:** 3 at par, 2 up to par × 1.2 rounded down, 1 above that.
- **Attempts:** every press of PLAY counts, even if stopped. The daily score is the number of the winning attempt; replaying afterwards improves ticks and stars, never attempts.
- **Hints:** after 3 failures, then after 6, on the same heist. The first shows the first third of the optimal plan as a ghost, the second shows two thirds. They appear in the share (💡).

Screens and interactions

The game fits in about ten screens, all in portrait and usable with one thumb. There is no title screen: the first launch opens the first level directly.

Screen	Content	Actions
First launch	Level T1, with an animated hand drawing the path. Consent banner at the bottom, which does not block play.	Draw, PLAY; OK or No thanks on the banner
Home	"Daily heist #3", its status (to do, or cleared in N attempts with its stars), the streak 🔥, the countdown to the next heist	Play, Share (if cleared), Archive (optional), Tutorial, Settings
Planning	Grid, patrols as arrowed dotted lines, cones at tick 0. Top bar: name, plan ticks and par, pauses left. Bottom bar: Clear, PLAY, Hint when available.	Draw, erase, place pauses, run
Execution	The plan plays out, with a tick counter	×2, Stop (back to the plan; the attempt still counts)

Screen	Content	Actions
Failure	Slow-motion replay of the last 4 ticks at quarter speed, zoom on the thief and whatever saw it, cone in red, written cause ("Spotted by the guard") and attempt number	Tap the screen: back to the intact plan, with a red marker on the fatal cell
Win	Attempts, ticks, par, stars, and the emoji line that will go into the share	Share (primary button), Improve my score, Home
Challenge card	"Alex cleared heist #3 in 2 attempts · 38 ticks. Your turn."	Take the challenge, Later
Comparison	You and your friend side by side: attempts, ticks, stars, hints, and a verdict	See their plan, Share my result, Home
Archive (optional)	Past days of the test week, with their status	Play a past day, with no effect on the streak
Settings	Language, sound, vibration, optional nickname, editable consent, privacy page, version	—

Path drawing

- Drawing starts from the thief, or from the last cell of the path to extend it. Touching another cell of the path cuts it there and continues from that cell.
- A cell is added only when the finger enters its central zone (70% of the cell): no stray jumps at cell borders.
- A fast swipe that skips cells is filled in cell by cell, in a straight line, up to the first wall.
- Moving back onto the second-to-last cell erases the last one. The Clear button empties the whole path.
- The path may revisit a cell already drawn, except through an immediate U-turn.
- Once the diamond is picked up, reaching the exit ends the drawing and makes PLAY pulse. Until then, PLAY stays greyed out, with a short instruction ("Grab the diamond, then get out").

- While drawing, the ghost preview shows guards, cameras and lasers at the tick of the cell under the finger. It is on by default, behind the `ghost` flag, so it can be tested without.
- Each new cell gives a short sound and, where the browser allows it, an 8 ms vibration.

Pauses

- A tap on a path cell, except the last one, adds a one-tick pause, shown as a badge (⏸ 2). A tap is a touch of less than 8 px and less than 250 ms.
- Tapping again increments. Once the level's maximum is reached, in total, the cell goes back to zero.
- A pause on the start cell makes the thief wait before leaving.

Hints and accessibility

- The hint shows as a dotted ghost path; the player can draw over it.
- Buttons of at least 44 px; text of at least 14 px; enough contrast in full sunlight.
- Danger is never signalled by colour alone: cones are tinted and outlined, failures come with an icon and text.
- If the system asks for reduced motion, the slow-motion replay is shortened and the screen does not shake.

Consent banner

"This prototype measures your games anonymously to decide whether it deserves to exist. No personal data." OK and No thanks buttons, plus a "Learn more" link to the privacy page.

Sharing and challenges

Sharing borrows Wordle's mechanics: a readable result that gives nothing away, sent in one tap, with a link that opens the same heist without any install.

What gets sent

```
Daily Heist #3 🕵️ cracked in 4 tries
👁️👁️📷✅ · 41 ticks ⭐⭐
Your turn: https://<domain>/c/3#AQME…
```

```
Casse du jour n°3 🕵️ réussi en 4 essais
👁️👁️📷✅ · 41 ticks ⭐⭐
À toi : https://<domaine>/c/3#AQME…
```

- One emoji per attempt: 👁️ guard, 📷 camera, 🔴 laser, ✅ success. Beyond 6 attempts: the first 5, "…", then ✅ and the total. One 💡 per hint used.
- Nothing reveals the solution. An image of the path would hand the day's answer to whoever receives it, so it is excluded. The link preview shows the heist map without any path, and the friend's plan is only revealed after the recipient's own win.

The gesture

- A single Share button, on the win screen and on Home once the heist is cleared.
- On mobile it opens the phone's share sheet directly. On desktop, or if the share sheet fails, the text is copied with a "Copied" message.
- Three variants to compare on real phones (story E5): text only with the link inside; text and link as separate fields; plus an image of the map. Depending on the app, the text, the link or the image may get dropped. The variant kept is the one that arrives intact in WhatsApp, Messages and Discord.

The link

- Format: `https://<domain>/c/<day>#<payload>`. The path gives the day and serves the preview (title, map image). The fragment after "#" carries the sender's result and is never sent to the server.
- Payload: version, day, attempts, ticks, stars, hints, failure causes, anonymous share ID, plan and optional nickname. The exact format is in the technical specs.
- The link always opens the heist of its own day, even if the recipient is already on the next day in their time zone.
- An unreadable payload simply opens the daily heist, without a challenge card.

Arrival

1. Challenge card: "Alex cleared heist #3 in 2 attempts · 38 ticks. Your turn." Without a nickname: "A friend cleared…".
2. If the recipient has never played: the three-level express tutorial (T1, T2, T4), about 90 seconds.
3. The linked heist, with the same rules.

4. Comparison: attempts first, then ticks. The friend's plan becomes visible after the recipient's win, or after the second hint.
5. "Share my result" sends a link back in turn: that is the rematch.

What we measure

Step	Event
Tap on Share	`share_tapped`
Share completed, or text copied	`share_result`
Link opened by another device	`link_opened`, with a sender ID different from the device's own
Challenge accepted	`challenge_cta` (play)
Linked heist cleared	`challenge_result`
New active player: at least 3 heists, or a return the next day	computed from the events

This is the funnel that will be used to compute K at gate 2: it must work from gate 1.

Content

Twelve levels in total: 5 tutorial levels, 3 of which form the express tutorial, and 7 daily heists, one per test day. Three of them are signature heists, built by hand around a specific moment. Target pars are design targets, not measurements.

Tutorial

Level	What it teaches	Elements	Target par	Success criterion
T1 "The plan"	Draw to the diamond, then to the exit, and run	No enemies	8 to 12 ticks	Cleared in under 60 s, without help

Level	What it teaches	Elements	Target par	Success criterion
T2 "The gaze"	A cone kills; go around it; read the slow-motion replay	1 static guard	10 to 14	The failure is understood from the first replay
T3 "The patrol"	Guards move: timing matters	1 patrolling guard	12 to 18	Cleared in 3 attempts or fewer
T4 "The pause"	A tap on the path makes the thief wait	1 guard, narrow passage, one mandatory pause	12 to 18	Pause found without help
T5 "The camera"	A rotating cone, and its blind spot	1 camera, 1 guard	14 to 20	Cleared in 4 attempts or fewer

The express tutorial, offered to new players arriving from a link, chains T1, T2 and T4.

Daily heists

Day	Type	Elements	Target par	Intent
D1	Normal	2 guards	16 to 22	Reassure: most players clear it in 1 to 3 attempts
D2	Normal	2 guards, 1 camera	18 to 24	Combine two known elements
D3	Signature "The U-turn"	2 back-and-forth guards	20 to 26	Slip behind a guard's back on the exact tick it turns around
D4	Normal	1 guard, 1 laser, with a help bubble on first encounter	18 to 24	Introduce the laser
D5	Signature "The blind spot"	1 camera, 1 laser, 1 guard	22 to 30	Wait under a camera while it sweeps the room
D6	Normal	3 entities	20 to 28	Leave several possible routes

Day	Type	Elements	Target par	Intent
D7	Signature "The changing of the guard"	2 guards crossing paths, 1 camera	24 to 32	A one-tick window, with a mandatory pause

What makes a signature heist

- A unique moment, written in one sentence before drawing, in the file's `moment` field.
- The optimal plan passes within one cell of a cone at least twice: this is the solver's tension metric.
- The solution can be deduced from what is visible and from the ghost preview, never from random trial and error.
- Two people play it before the test. If neither reacts at the intended moment, it gets reworked.

Quality bar for every heist

- Validated by the solver, with a par inside the target.
- 4 entities at most: readable in 3 seconds.
- At least two plausible routes at the start, so the player thinks instead of following a corridor.
- No invisible trap: everything that kills is visible at tick 0 (patrols as dotted lines, laser cycles shown).
- Drawn by hand, possibly from a generator draft, then tuned in the viewer until it passes this bar.

Metrics and test protocol

Each gate 1 indicator has an exact definition. It is computed on the testers who accepted measurement, excluding internal devices.

Events

Event	When	Properties
`app_open`	Launch, or return to the foreground after 30 min	entry (direct, link, portal), is_first_open, link_day, sender_share_id
`consent_set`	Answer to the banner (sent only if yes)	granted
`level_start`	Start of a level	level_id, mode (tutorial, daily, archive, challenge), day
`attempt_start`	Press on PLAY	level_id, attempt, plan_ticks, pauses_used
`attempt_fail`	Failure	level_id, attempt, cause (guard, camera, laser), tick, ms_since_level_start
`attempt_win`	Win	level_id, attempt, ticks, par, stars, hints_used, ms_since_level_start, first_win
`hint_used`	Hint shown	level_id, hint_index, attempt
`share_tapped`	Press on Share	context (win, home, comparison), day
`share_result`	End of the share	method (native, clipboard), outcome (completed, cancelled, error), variant
`link_opened`	Arrival from a link carrying a payload	day, sender_share_id, is_new_player, is_self
`challenge_cta`	Choice on the challenge card	action (play, later)
`challenge_result`	End of the received heist	day, beat_sender, delta_attempts, delta_ticks
`friend_plan_viewed`	Friend's plan watched	day

Event	When	Properties
`streak_day`	Win on the daily heist, on the same day	streak
`heartbeat`	Every 30 s of active play	active_ms
`error`	JavaScript error	short message, scene

Super-properties attached to every event: version, build, language, device (mobile or desktop), recruitment source (`src`), active flags, internal marker. Active time counts while the game is in the foreground and the player has interacted in the last 30 seconds.

Indicator definitions

Indicator	Exact definition	Kill threshold
Clarity	Share of testers whose first win on T1 comes less than 60 s after its `level_start`, without a hint	Below 60%
Voluntary play	Share of testers who, on the local day of their first launch, accumulate at least 8 minutes of active time or complete the tutorial and the daily heist	Below 40%
Retry after failure	Share of `attempt_fail` events followed by an `attempt_start` on the same level within 2 minutes	Below 60%
Spontaneous sharing	Share of testers with at least one completed or copied `share_result` during the week	Below 10%
Next-day return	Share of testers with an `app_open` on the local day after their first launch, among those who arrived before day 7	Below 30%

The daily format limits day-one content: a fast player who finishes everything in 6 minutes must not count as a bored player. Hence the "or" in the voluntary play definition.

Also tracked, without a threshold: the share funnel, failure causes per level, the distribution of attempts per heist, hint usage, and results on signature days compared with the others.

Feel sessions

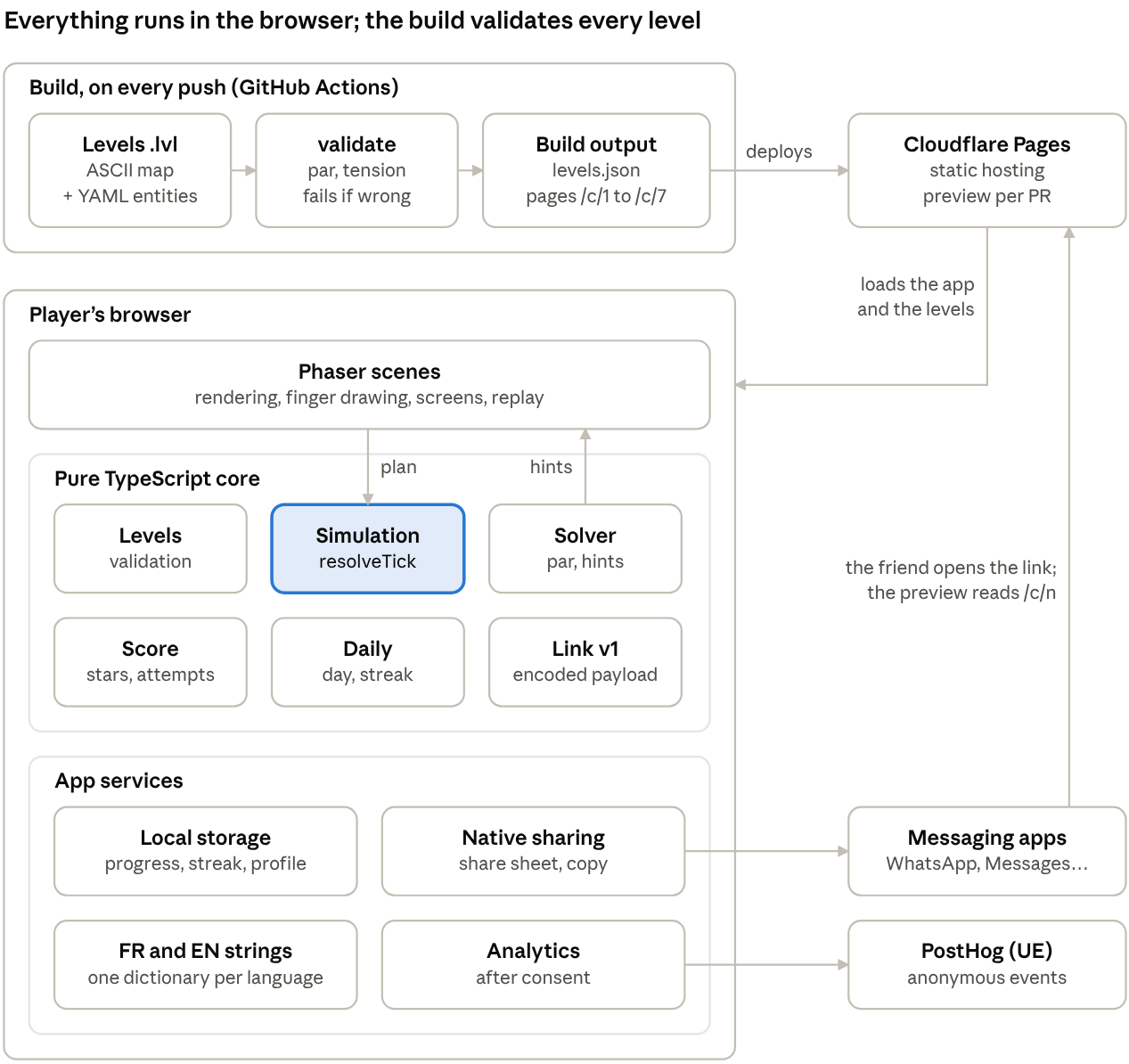
- Two 30-minute sessions with three people each, in person or by video call with the phone screen shared. Players think aloud; nobody helps them.
- Session 1: T1 to T5 and two heists, ghost preview on. Session 2, after fixes: half of the players without the ghost preview.
- After each session, issues are ranked (blocking, annoying, minor); blocking issues are fixed before the test week.

Moment	What to watch	Warning sign
First drawing	Do they draw without any explanation?	Hesitates more than 10 s
Fast drawing	Skipped cells, accidental erasing	More than one accidental erase per level
Reading cones	Can they say why they failed?	"I didn't see it" more than once
Failure	Do they retry right away?	Sighs or puts the phone down without retrying
Pause	Do they discover the tap on T4?	Cannot find it without help
Sharing	"Who would you send this to? Show me."	Cannot find the button, or sees nobody to send it to

Test week

1. Recruit 20 to 30 testers: about 15 French-speaking friends and acquaintances, 5 to 10 players from r/playmygame and puzzle Discord servers, plus Poki Playtesting if access is granted. Each group gets a tagged link (`?src=…`).
2. Send a single message in the tester's language, without any mention of sharing: "A little heist game, a new one every day for 7 days. Play whenever you like."
3. Do not remind anyone during the 7 days: no message, no nudge.
4. Day 8: a 5-question survey (fun from 1 to 5; what frustrated you; would you keep playing for another month; did you show it to anyone, and how; what would have made you want to share it).
5. Decide with the table in Framing, and write it up on one page.

Technical architecture



Architecture · build, browser and external services

The TypeScript core depends on neither the DOM nor randomness: the game, the solver, the viewer and the `validate` tool all call the same `resolveTick` function. The only outgoing traffic is the events sent after consent and the links the player shares themselves.

Technical specs

Everything runs in the browser: a pure, deterministic TypeScript core, shared by the game, the solver and the tools, under a Phaser rendering layer. There is no application server.

Stack

Component	Choice	Reason
Language	TypeScript in strict mode	One language for the game, the tools and the tests
Engine	Phaser 4, exact version pinned	Choice made in the decision dossier; 2D WebGL rendering
Tooling	Vite, ESLint, Prettier, Vitest	Fast start; core tests without a browser
Hosting	Cloudflare Pages, static files	Preview per branch, no server to manage
Analytics	PostHog, hosted in Europe, behind an adapter	Funnels, retention and cohorts without building them; replaceable
Sounds	6 royalty-free sounds (CC0, e.g. Kenney)	No cost, no licence to track
Continuous integration	GitHub Actions	Tests, level validation, build, deployment

Repository layout

```
src/
  core/      pure rules: level, sim, vision, solver, score, plan, daily,
payload
  game/      Phaser: scenes, rendering, path drawing, effects
  app/       router, storage, analytics, consent, sharing, i18n, config
levels/
  tutorial/  t1.lvl … t5.lvl
  daily/     d1.lvl … d7.lvl
tools/       validate.ts, og.ts, gen.ts (optional)
tests/       unit, golden, payload, daily, e2e
public/      icons, sounds
```

The `core/` folder imports neither Phaser nor the DOM: it runs identically in the browser, in Node and in the tests.

Level format

One `.lvl` file per level: an ASCII map plus entities in YAML. Here is a format example, which is not one of the test levels:

```
name: Example
type: normal        # normal | signature | tutorial
moment: ""          # one sentence, mandatory for a signature heist
maxPauses: 3
map: |
  ########
  #S.....#
  #.####.#
  #......#
  ##.##.##
  #...L..#
  #.####.#
  #......#
  ###.####
  #......#
  #.....E#
  ########
guards:
  - id: g1
    route: [[1,3], [2,3], [3,3], [4,3], [5,3]]
    mode: pingpong  # loop | pingpong ; 5 cells back and forth = 8 ticks
    phase: 0
    range: 3
cameras:
  - id: c1
    at: [2,8]       # a wall cell
    dirs: [S, E]
    hold: 3         # 3 ticks per direction = 6-tick cycle
    phase: 0
    range: 3
lasers:
  - id: l1
```

```
    cells: [[3,9], [4,9], [5,9]]
    on: 2
    off: 2
    phase: 1
```

- `map`: 12 lines of 8 characters. `#` wall, `.` floor, `S` start, `L` diamond (loot), `E` exit, each exactly once.
- `guards[].route`: adjacent cells. In `loop` mode the last cell touches the first; in `pingpong` mode the guard goes out and back without repeating the endpoints. A repeated cell is a wait, where `face` can set the facing. `phase` shifts the start. A static guard has a single cell and a `face`.
- `cameras[]`: `at` on a wall; `dirs` among N, E, S, W; `hold` ticks per direction; `phase`.
- `lasers[]`: aligned, contiguous floor cells; `on`, `off`, `phase`.
- The build rejects any file where a period does not divide 24, where the start is seen at tick 0, or that the solver cannot clear within 64 ticks.

Simulation

- `run(level, plan)` returns one frame per tick (positions, facings, seen cells, active lasers), then the outcome, the cause, the tick and the pickup moment.
- A single function, `resolveTick(state, step)`, is called by the game, the solver and the viewer: there are never two implementations of the rules.
- `visibleCells(level, origin, facing, range)` is cached per entity and per tick modulo the period.
- `core/` uses integers only: no `Math.random`, no `Date`, no floats, enforced by a dedicated ESLint rule.
- Line of sight: integer Bresenham, endpoints excluded. A diagonal gap between two walls lets sight through; levels avoid it.

```
function lineOfSight(x0: number, y0: number, x1: number, y1: number,
                     isWall: (x: number, y: number) => boolean): boolean {
  const dx = Math.abs(x1 - x0), dy = -Math.abs(y1 - y0)
  const sx = x0 < x1 ? 1 : -1, sy = y0 < y1 ? 1 : -1
  let err = dx + dy, x = x0, y = y0
  while (true) {
    const e2 = 2 * err
    if (e2 >= dy) { err += dy; x += sx }
    if (e2 <= dx) { err += dx; y += sy }
    if (x === x1 && y === y1) return true
    if (isWall(x, y)) return false
  }
}
```

Solver

- Breadth-first search over the state (x, y, last direction, t modulo P, pauses left, diamond taken): at most 96 × 5 × 24 × 4 × 2, about 92,000 states.
- Transitions: N, E, S or W onto a floor cell, except an immediate U-turn; a wait if a pause is left. Every transition goes through `resolveTick`; a state where the thief is caught is dropped. Maximum depth: 64 ticks.
- Outputs: par; canonical optimal plan (order N, E, S, W, wait); minimum pauses; number of optimal plans, capped at 1,000; tension, i.e. the number of ticks of the optimal plan where the thief is one orthogonal cell away from a seen cell.
- Uses: validation at build time, displayed par, hints, viewer, and later verification of received plans.

Rendering and input

- One Phaser scene per screen (Boot, Home, Level, Result, Challenge, Settings). The static grid is drawn once into a texture.
- Simple vector shapes: no illustrations at gate 1.
- Rendering interpolates between two simulation frames and never computes a rule.
- Path drawing: pointer events, a single finger; 70% central zone; tap under 8 px and 250 ms; skipped cells filled in one by one. These thresholds are constants, tunable during feel sessions.
- Portrait 9:16 frame, centred, at most 480 px wide on desktop. Cells fill the space between a 64 px top bar and a 96 px bottom bar.

Daily loop and configuration

- Day = number of local calendar days between `TEST_START_DATE` and today, plus 1, clamped to 1–7. Before the start: tutorial and countdown. After day 7: archive and an end message.
- The streak only grows on a win of the daily heist on the same day; a day without a win resets it.
- `config.ts` holds the start date, the domain and the flags (`ghost`, on by default). A flag can be forced through the URL (`?flag=ghost:0`), is remembered in the profile, and is sent with every event.

Link payload (v1)

Bytes	Field
0	Version (1)
1	Day (1 to 255)
2	Winning attempt (0 = not cleared yet)
3	Best ticks
4	Stars (2 bits), hints (2 bits), number of causes (4 bits)
5–6	Causes of the first 8 failures, 2 bits each
7–10	Share ID, 32 random bits
11	Plan length n (64 at most)
12 to 11 + ⌈3n/8⌉	Plan, 3 bits per step: N, E, S, W, wait
Next	Nickname length, then the nickname in UTF-8 (12 bytes at most)
Last	Checksum: XOR of all previous bytes

The whole payload is base64url-encoded; a full link is under 200 characters. At launch, the app reads the fragment, stores it, then removes it from the address bar (`history.replaceState`), so no tool can capture it afterwards.

Sharing and previews

- `navigator.share` when available, otherwise `navigator.clipboard.writeText`. An `AbortError` means the player cancelled.
- Pages `/c/1` to `/c/7` are generated at build time: the same app, with its `og:title`, `og:description` and `og:image` tags. The 1200 × 630 image shows the map without any path; it is drawn as SVG then converted to PNG (resvg).
- The root `/` serves the local daily heist.

Local storage

Key	Content
`pc1:profile`	anonId, shareId, nickname, language, consent, internal marker, tutorial done, flags, schema version
`pc1:days`	Per day: attempts, winning attempt, causes, best ticks, stars, hints, encoded plan
`pc1:levels`	Tutorial progress
`pc1:streak`	Current streak, best streak, last day won
`pc1:inbox`	Received challenges: sender, day, payload

All access goes through a module that catches errors (private browsing, full storage) and falls back to memory.

Analytics and privacy

- A single entry point, `track(event, props)`. Nothing leaves before consent; a refusal means no data, not degraded data.
- PostHog: autocapture and session recording disabled; identifier = anonId.
- Internal testers flagged with `?internal=1`, then excluded from the indicators.
- No personal data. The nickname only leaves the device inside links the player shares themselves.
- Data deleted at the end of gate 2. To be reviewed before any wider public test.

Performance and compatibility

- First playable screen in under 3 s on 4G on a mid-range Android, with less than 3 MB transferred. CrazyGames recommends under 10 s and under 20 MB (CrazyGames); the target is tighter because a shared link must open right away.
- 60 frames per second in play.
- Browsers: Safari on iOS 16 and later, recent Chrome on Android, and common desktop browsers.

Tests

- Unit: every rule in the Game rules section (cone per facing, line of sight, patrols, cameras, lasers, swap, detection order, win).
- Golden: for every level, simulating the optimal plan wins in exactly "par" ticks.
- Replay: a winning plan, encoded into a payload and then decoded, gives the same outcome and the same ticks.
- Payload: round trip on 10,000 random payloads.
- Daily: correct local day around daylight saving changes (Europe/Paris, America/Chicago).
- End-to-end (Playwright, two scenarios): first launch then T1 cleared with an injected plan; Share button copying the text on desktop.

Backlog

52 hours across 34 stories and 8 epics, plus two optional items to cut first. This fine-grained breakdown slightly exceeds the 50-hour ceiling of the decision dossier: gate 1 runs with no margin.

Epics

Epic	Hours	Done when
A — Foundations	3	An empty PR passes CI and deploys to preview
B — Simulation core	8.5	The rules are tested and every level is validated at build time
C — Game	12	You draw, play, fail with a replay, and win with stars

Epic	Hours	Done when
D — Daily loop	2.75	The heist changes at midnight and the streak is counted
E — Sharing and challenges	5.25	A shared link opens the heist for a friend and compares scores
F — Content	9	All 12 levels pass the quality bar
G — Measurement	3	The five indicators can be computed
H — Feel and testing	8.5	The gate 1 decision is written down
Total	52	Plus 1.75 h of optional items

Stories

ID	Story	Role	Hours	Depends on	Acceptance criteria	Status
A1	Repository and tooling: Vite, strict TypeScript, Phaser 4, ESLint, Vitest, CI, Cloudflare Pages deployment with a preview per PR	Dev	1.5	—	Green CI on an empty PR; one preview URL per PR; empty scene runs smoothly on a mid-range Android	To do
A2	Configuration, safe local storage, anonymous IDs, URL flags, internal marker	Dev	0.75	A1	No crash in private browsing; `?flag=ghost:0` and `?internal=1` remembered	To do
A3	FR and EN strings: typed dictionaries, language detection, choice in settings	Dev	0.75	A1	No hard-coded string; language switch without reload	To do
B1	Level format, parser and validator	Dev	1.5	A1	The example parses; every invalid file returns a precise error with its line	To do
B2	Simulation: patrols and facings, cones and line of sight, cameras, lasers, resolution order, outcome and cause	Dev	3.5	B1	One test per rule in the Game rules section, including the swap and wall cases	To do

ID	Story	Role	Hours	Depends on	Acceptance criteria	Status
B3	Solver: par, canonical optimal plan, minimum pauses, number of optimal plans, tension	Dev	2.5	B2	Simulating the optimal plan wins in exactly "par" ticks; under 50 ms per level	To do
B4	`validate` CLI and `levels.json` compilation, wired into CI	Dev	1	B3	CI fails on an unsolvable or malformed level, or one whose period does not divide 24	To do
C1	Rendering: grid, entities, cones as exact cells, dotted patrols, scaling	Dev	3	B2	Tinted cells are exactly those of `visibleCells`; readable on a 360 × 640 screen	To do
C2	Path drawing: central zone, erase by going back, cut, pauses on tap, 64-step limit, auto-finish at the exit	Dev	3	C1	Rules in the Screens and interactions section; 20 fast drawings without an accidental erase	To do
C3	Execution: playback of simulation frames at 4 ticks/s, ×2, Stop	Dev	1.5	C2	Displayed positions = simulated positions at every tick (dev assertion)	To do
C4	Failure: slow-motion replay of the last 4 ticks, zoom, written cause, back to the intact plan with a marker	Dev	1.5	C3	Retry in one tap; path and pauses kept	To do
C5	Win: attempts, ticks, par, stars, emoji line, Share, Improve, Home	Dev	1	C3	Correct stars; replaying after the win does not change attempts	To do
C6	Ghost preview, behind the `ghost` flag	Dev	1	C2	While drawing, guards and cones are shown at the tick of the cell under the finger	To do
C7	Hints after 3 then 6 failures	Dev	0.5	B3, C2	First third then two thirds of the optimal plan as a ghost; 💡 counted in the share	To do
C8	Minimal sounds and vibration, toggleable	Dev	0.5	C3	6 royalty-free sounds; vibration only where the browser supports it; settings remembered	To do

ID	Story	Role	Hours	Depends on	Acceptance criteria	Status
D1	Daily logic: start date, local day, countdown, streak, before and after the week	Dev	1.5	A2	Tests around daylight saving changes; the streak only counts same-day wins	To do
D2	Home and flow: first launch on T1, return to Home, full tutorial	Dev	1.25	D1, C5	No title screen on first launch; a known player lands on Home	To do
E1	Payload v1: encoding, base64url, version, checksum	Dev	1	B2	Round trip on 10,000 random payloads; link under 200 characters	To do
E2	Share button: share sheet, three variants, copy fallback, measured funnel	Dev	1.25	E1, C5	One tap from the win screen to the share sheet; "Copied" on desktop	To do
E3	Pages `/c/1` to `/c/7` and preview images generated at build time	Dev	0.75	B4	A link pasted in WhatsApp or Messages shows the title and the map	To do
E4	Link arrival: challenge card, express tutorial, comparison, friend's plan after the recipient's own win	Dev	1.5	E1, D2	The friend's plan stays hidden until the recipient has won or taken the second hint	To do
E5	Share matrix on real phones and choice of variant	Dev, PM	0.75	E2, E3	Table filled in (iOS and Android × WhatsApp, Messages, Discord, Messenger, Instagram); default variant chosen	To do
F1	Dev viewer: level, cones at a chosen tick, optimal plan replayed, metrics	Dev	1.5	B3, C1	Any `.lvl` file opens in `/dev/viewer`	To do
F2	5 tutorial levels	Design	1.5	F1	Each level teaches a single idea; T1 cleared in under 60 s by 3 people out of 3	To do
F3	7 daily heists, 3 of them signature heists, each with its "moment" sentence	Design	6	F1	Quality bar from the Content section; each heist played by 2 people	To do

ID	Story	Role	Hours	Depends on	Acceptance criteria	Status
G1	Analytics: PostHog adapter, super-properties, events, active time	Dev	1.25	A2	Every event in the table arrives with its properties; no URL with a fragment	To do
G2	Consent and privacy page	Dev	0.5	G1	Nothing sent before OK; a refusal = zero events	To do
G3	Gate 1 dashboard: five indicators, share funnel, failure causes	PM	1.25	G1	Each indicator computed per its definition, on test data	To do
H1	Feel session #1	PM	1.5	C1 to C5, F2, 3 heists from F3	Issues logged and ranked by severity	To do
H2	Feel fixes	Dev	2	H1	Blocking issues fixed	To do
H3	Session #2, with and without the ghost preview, then final fixes	PM, Dev	1.5	H2	Default value of `ghost` chosen	To do
H4	Recruitment: 20 to 30 testers, message with no mention of sharing, tagged links, Poki request	PM	1.5	—	List ready before day 1	To do
H5	Test week with no reminders, then the survey	PM	1	H4, G3	No reminder during the 7 days	To do
H6	Gate 1 decision: indicators filled in, one-page note	PM	1	H5	Written decision: stop, fix the feel, pivot, or launch the cold test	To do
O1	Optional: playable archive	Dev	0.75	D2	Past days are playable without touching the streak	To do
O2	Optional: draft generator	Dev	1	B3	Drafts filtered by par and by tension	To do

Suggested order

In weeks of evenings of about 17 hours:

1. **Week 1:** A1 to A3, B1 to B4, then C1. The core is tested before any polished rendering.
2. **Week 2:** C2 to C8, D1, D2, E1, E2, F1; start recruitment (H4) in parallel.
3. **Week 3:** E3 to E5, G1 to G3, F2, F3, then the feel sessions (H1 to H3).

4. **Week 4:** test week (H5); day 8, decision (H6).

Cut order

If time slips, cut in this order: O2, O1, C8, then the second hint of C7. Never cut C2 (path drawing), C4 (failure replay), E2 (one-tap sharing), F3 (signature heists) or H1 to H3 (feel): they are the possible sources of a false negative.

Guidelines for a coding agent

A coding agent can start with this document alone. It codes in backlog order, one story at a time, and stops to ask as soon as a rule is missing.

Method

1. Read Game rules, Screens and interactions, Technical specs and Backlog before writing the first line of code.
2. Take stories in the suggested order, respecting dependencies. One story = one branch = one PR, titled for example "[B2] Simulation".
3. Write the tests that express the acceptance criteria first, then the code.
4. Set the story to "In progress" when starting, then to "Done" once the definition of done is met.

Definition of done

- Acceptance criteria met and listed in the PR.
- Tests added; green CI (types, lint, tests, level validation, build).
- UI story: preview URL checked on a real phone.
- Measurement story: events seen arriving in the analytics tool.
- Changed behaviour: the relevant section of this document is updated in the same PR.

Never without approval

- Change a game rule. Rules live in `resolveTick` and `visibleCells`, shared by the game and the solver; a change goes through a failing test first, then through the Game rules section.
- Introduce randomness, the clock or floats into `core/`.
- Edit `levels.json` by hand: edit the `.lvl` files, then rerun `validate`.

- Add a dependency without justifying its weight (3 MB budget).
- Send an event without going through `track`, any personal data, or a URL with its fragment.
- Write displayed text anywhere other than the FR and EN dictionaries.
- Go out of scope: editor, ads, accounts, server, notifications.

When in doubt

Pick the simplest option that respects the rules, note it as an open question in the PR, and never invent a mechanic. Anything to be tested in a feel session goes behind a flag.

Decisions to make before starting

Eight decisions block specific stories. Each has a default value, to be confirmed by Jeremy.

Decision	Default	Blocks
Domain name	Short, .com or .app, about $15 to $20 a year	E3, E5
Start date of the test week	A Monday, about 3 weeks after development starts	D1, H4
Analytics tool	PostHog, hosted in Europe	G1
Displayed game name	"Perfect Crime", provisional; check trademarks before the stores	E3
Nickname in links	Optional, empty by default ("A friend")	E1, E4
Tester groups	About 15 French-speaking friends, 5 to 10 English-speaking players, Poki if accepted	H4
PM role	Jeremy: feel sessions, recruitment, decision	H1, H4
Poki Playtesting access	Request sent in week 1; lead time unknown	H4

Starting points that the feel sessions may change: speed of 4 ticks per second, 70% central zone, tap thresholds, cone range, ghost preview on.